RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 1

I:\MEPC\63\23.doc
ANNEX 8

RESOLUTION MEPC.212(63)

Adopted on 2 March 2012

2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPS

THE MARINE ENVIRONMENT PROTECTION COMMITTEE,

RECALLING article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Envi ronment Protection Committee (the Committee) conferred upon it by international conventions for the prevention and control of marine pollution,

RECALLING ALSO that, at its sixty-second session, the Committee adopted, by resolution MEPC.203(62), amendments to the annex of the Pr otocol of 1997 to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (inclusion of regulations on energy efficiency for ships in MARPOL Annex VI),

NOTING the amendments to MARPOL Annex VI adopted at its sixty-second session by

of 1978 relating thereto (inclusion of regulations on energy efficiency for ships in MARPOL Annex VI),

NOTING the amendments to MARPOL Annex VI adopted at its sixty-second session by inclusion of a new chapter 4 for regulations on energy efficiency for ships, are expected to enter into force on 1 January 2013 upon their acceptance on 1 July 2012,

NOTING ALSO that regulation 20 (Attai ned EEDI) of MARPOL Annex VI, as amended, requires that the Energy Efficiency Design Index shall be calculated taking into account the guidelines developed by the Organization,

RECOGNIZING that the amendments to MARPOL Annex VI requires the adoption of relevant guidelines for smooth and uniform implem entation of the regulations and to provide sufficient lead time for industry to prepare,

HAVING CONSIDERED, at its sixty-third session, the draft 2012 Guidelines on the method of calculation of the attained Energy Efficiency Design Index (EEDI) for new ships,

1. ADOPTS the 2012 Guidelines on the method of calculation of the attained Energy Efficiency Design Index (EEDI) for new ships, as set out at annex to the present resolution;

1. ADOPTS the 2012 Guidelines on the method of calculation of the attained Energy Efficiency Design Index (EEDI) for new ships, as set out at annex to the present resolution;

2. INVITES Administrations to take the annexed Guidelines into account when developing and enacting national laws which give fo rce to and implement provisions set forth in regulation 20 of MARPOL Annex VI, as amended;

3. REQUESTS the Parties to MARPOL Annex VI and other Member Governments to bring the annexed Guidelines related to the Energy Efficiency Design Index (EEDI) to the attention of shipowners, ship operators, shipbuilders, ship designers and any other interested groups;

4. AGREES to keep these Guidelines under review in light of the experience gained; and

5. REVOKES the Interim Guidelines circulated by MEPC.1/Circ.681, as from this date.

RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 2

I:\MEPC\63\23.doc
ANNEX

2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPS

I:\MEPC\63\23.doc
ANNEX

2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPS

CONTENTS

1 Definitions

2 Energy Efficiency Design Index (EEDI) including the equation

2.1 CF ; conversion factor between fuel consumption and CO2 emission
2.2 Vref ; ship speed
2.3 Capacity
2.3.1 Bulk carriers, tankers, gas carriers, ro-ro cargo ships and general cargo ships
2.3.2 Passenger ships and ro-ro passenger ships
2.3.3 Containerships
2.4 Deadweight
2.5 P ; Power of main and auxiliary engines
2.5.1 PME ; power of main engines
2.5.2 PPTO ; shaft generator
2.5.3 PPTI ; shaft motor
2.5.4 Peff ; output of innovative mechanical energy efficient technology
2.5.5 PAEeff ; auxiliary power reduction
2.5.6 PAE ; power of auxiliary engines
2.6 Vref, Capacity and P
2.7 SFC ; Specific fuel consumption
2.8 fj ; Correction factor for ship specific design elements
2.8.1 fj ; ice-class ships
2.8.2 fj ; shuttle tankers
2.8.3 fj ; other ship types
2.9 fw ; Weather factor

2.6 Vref, Capacity and P
2.7 SFC ; Specific fuel consumption
2.8 fj ; Correction factor for ship specific design elements
2.8.1 fj ; ice-class ships
2.8.2 fj ; shuttle tankers
2.8.3 fj ; other ship types
2.9 fw ; Weather factor
2.10 feff ; Availability factor of innovative energy efficiency technology
2.11 fi ; Capacity factor
2.11.1 fi ; ice-class ships
2.11.2 fi ; ship specific voluntary structural enhancement
2.11.3 fi ; bulk carriers and oil tankers under Common Structural Rules (CSR)
2.11.4 fi ; other ship types
2.12 fc ; Cubic capacity correction factor
2.12.1 fc ; chemical tankers
2.12.2 fc ; LNG carriers
2.13 Lpp ; Length between perpendiculars

APPENDIX 1 A generic and simplified power plant

APPENDIX 2 Guidelines for the development of electric power tables for EEDI (EPT-EEDI)

RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 3

I:\MEPC\63\23.doc
1 Definitions

MARPOL means the International Convention for the Prevention of Pollution from

1 Definitions

MARPOL means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto, as amended.

For the purpose of these Guidelines, the definitions in "REGULATIONS ON ENERGY EFFICIENCY FOR SHIPS" (RESOLUTION MEPC. 203(62)) apply.

2 Energy Efficiency Design Index (EEDI)

The attained new ship Energy Efficiency Design Index (EEDI) is a measure of ships energy efficiency (g/t*nm) and calculated by the following formula:

ref wc i
neff
i
MEFME i eff i effAEFAE
nPTI
i
neff
i
i AEeff i effi PTI
n
j
jAEFAE AEi MEi FME
nME
i
i ME
n
j
j
V f Capacity f f
SFC C P fSFC C P f P fSFC C PSFC C P f

1) ( ) (11) ( ) () (1) () (1) (1

* If part of the Normal Maximum Sea Load is provided by shaft generators, SFCME and CFME may for that part of the power be used instead of SFCAE and CFAE
** In case of P PTI(i)>0, the average weighted value of (SFCME . CFME) and

* If part of the Normal Maximum Sea Load is provided by shaft generators, SFCME and CFME may for that part of the power be used instead of SFCAE and CFAE
** In case of P PTI(i)>0, the average weighted value of (SFCME . CFME) and (SFCAE . CFAE) to be used for calculation of Peff

Note: This formula may not be able to apply to diesel-electric propulsion, turbine propulsion or hybrid propulsion system.

Where:

.1 CF is a non-dimensional conversion factor between fuel consumption measured in g and CO 2 emission also measured in g based on carbon content. The subscripts MEi and AEi refer to the main and auxiliary engine(s) respectively. CF corresponds to the fuel used when determining SFC listed in the applicable test report included in a Technical File as defined in paragraph 1.3.15 of NO x Technical Code ("test report included in a NO x technical file" hereafter). The value of CF is as follows:

Type of fuel Reference Carbon content CF (t-CO2/t-Fuel)
1 Diesel/Gas Oil ISO 8217 Grades DMX through DMB 0.8744 3.206
2 Light Fuel Oil (LFO) ISO 8217 Grades RMA through RMD 0.8594 3.151
3 Heavy Fuel Oil
(HFO)
ISO 8217 Grades RME through

Type of fuel Reference Carbon content CF (t-CO2/t-Fuel)
1 Diesel/Gas Oil ISO 8217 Grades DMX through DMB 0.8744 3.206
2 Light Fuel Oil (LFO) ISO 8217 Grades RMA through RMD 0.8594 3.151
3 Heavy Fuel Oil
(HFO)
ISO 8217 Grades RME through
RMK 0.8493 3.114
4 Liquefied Petroleum
Gas (LPG)
Propane 0.8182 3.000
Butane 0.8264 3.030
5 Liquefied Natural
Gas (LNG) 0.7500 2.750
RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 4

I:\MEPC\63\23.doc
.2 Vref is the ship speed, measured in nautical miles per hour (knot), on deep water in the condition corresponding to the Capacity as defined in paragraphs 2.3.1 and 2.3.3 (in case of passenger ships and ro-ro passenger ships, this condition should be summer load draught as provided in paragraph 2.4) at the shaft power of the engine(s) as defined in paragraph 2.5 and assuming the weather is calm with no wind and no waves.

.3 Capacity is defined as follows:

.1 For bulk carriers, tankers, gas tankers, ro-ro cargo ships, general

paragraph 2.5 and assuming the weather is calm with no wind and no waves.

.3 Capacity is defined as follows:

.1 For bulk carriers, tankers, gas tankers, ro-ro cargo ships, general cargo ships, refrigerated cargo carrier and combination carriers, deadweight should be used as Capacity.

.2 For passenger ships and ro-ro passenger ships, gross tonnage in accordance with the International Convention of Tonnage Measurement of Ships 1969, Annex I, regulation 3 should be used as Capacity.

.3 For containerships, 70 per cent of the deadweight (DWT) should be used as Capacity. EEDI values for containerships are calculated as follows:

.1 attained EEDI is calculated in accordance with the EEDI formula using 70 per cent deadweight for Capacity.

.2 estimated index value in the Guidelines for calculation of the reference line is calculated using 70 per cent deadweight as:

ref
NME
i
AEi ME
V
PP
Value IndexEstimated

DWT % 70
215190
1144 . 3 1

.3 parameters a and c for containerships in Table 2 of regulation 21 of MARPOL Annex VI are determined by plotting the estimated index value against 100 per cent

i
AEi ME
V
PP
Value IndexEstimated

DWT % 70
215190
1144 . 3 1

.3 parameters a and c for containerships in Table 2 of regulation 21 of MARPOL Annex VI are determined by plotting the estimated index value against 100 per cent deadweight i.e. a=174.22 and c=0.201 were determined.

.4 required EEDI for a new containership is calculated using 100 per cent deadweight as:

Required EEDI = (1-X/100) · a · 100% deadweight c

Where X is the reduction factor (in percentage) in accordance with Table 1 in regulation 21 of MARPOL Annex VI relating to the applicable phase and size of new containership.

.4 Deadweight means the difference in tonnes between the displacement of a ship in water of relative density of 1,025 kg/m 3 at the summer load draught and the lightweight of the ship. The summer load draught should be taken as the maximum summer draught as certified in the stability booklet approved by the Administration or an organization recognized by it.
RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 5

.5 P is the power of the main and auxiliary engines, measured in kW. The subscripts ME and AE refer to the main and auxiliary engine(s), respectively. The summation on i is for all engines with the number of engines (nME). (See diagram in appendix 1.)

.1 PME(i) is 75 per cent of the rated installed power (MCR *) for each main engine (i).

The influence of additional shaft power take off or shaft power take in is defined in the following paragraphs.

.2 Shaft generator

In case where shaft generator(s) are installed, PPTO(i) is 75 per cent of the rated electrical output power of each shaft generator.

For calculation of the effect of shaft generators two options are available:

Option 1:

.1 The maximum allowable deduction for the calculation of PME(i) is to be no more than PAE as defined in paragraph 2.5.6. For this case, PME(i) is calculated as:

PME(i) =) () (75 . 0 i PTOi ME PMCR
or

Option 2:

Option 1:

.1 The maximum allowable deduction for the calculation of PME(i) is to be no more than PAE as defined in paragraph 2.5.6. For this case, PME(i) is calculated as:

PME(i) =) () (75 . 0 i PTOi ME PMCR
or

Option 2:

.2 Where an engine is installed with a higher rated power output than that which the propulsion system is limited to by verified technical means, then the value of PME(i) is 75 per cent of that limited power for determining the reference speed, Vref and for EEDI calculation.

* The value of MCR specified on the EI APP certificate should be used for calculation. If the main engines are not required to have an EIAPP certificate, the MCR on the nameplate should be used.
RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 6

I:\MEPC\63\23.doc
The following figure gives guidance for determination of PME(i) :

.3 Shaft motor

In case where shaft motor(s) are installed, PPTI(i) is 75 per cent of

The following figure gives guidance for determination of PME(i) :

.3 Shaft motor

In case where shaft motor(s) are installed, PPTI(i) is 75 per cent of the rated power consumption of each shaft motor divided by the weighted average efficiency of the generator(s).

The propulsion power at which Vref is measured, is:

Shaft i PTIi ME P P), () (

Where:

Geni PTI i PTIShaft i PTI PP) ( ) (), (

) (i PTI is the efficiency of each shaft motor installed
Gen is the weighted average efficiency of the generator(s)

Where the total propulsion power as defined above is higher than 75 per cent of the power the propulsion system is limited to by verified technical means, then 75 per cent of the limited power is to be used as the total propulsion power for determining the reference speed, Vref and for EEDI calculation.

In case of combined PTI/PTO, the normal operational mode at sea

In case of combined PTI/PTO, the normal operational mode at sea will determine which of these to be used in the calculation.

Note: The shaft motor's chain efficiency may be taken into consideration to account for the energy losses in the equipment from the switchboard to the shaft motor, if the chain efficiency of the shaft motor is given in a verified document.

.4 Peff(i) is the output of the innovative mechanical energy efficient technology for propulsion at 75 per cent main engine power.

Mechanical recovered waste energy directly coupled to shafts need not be measured, since the effect of the technology is directly reflected in the Vref.

In case of a ship equipped dual-fuel engine or a number of engines, the CFME and SFC ME should be the power weighted average of all the main engines.

.5 PAEeff (i) is the auxiliary power reducti on due to innovative electrical energy efficient technology measured at PME(i).

.6 PAE is the required auxiliary engine power to supply normal

average of all the main engines.

.5 PAEeff (i) is the auxiliary power reducti on due to innovative electrical energy efficient technology measured at PME(i).

.6 PAE is the required auxiliary engine power to supply normal maximum sea load including necessary power for propulsion machinery/systems and accommodation, e.g. main engine pumps, navigational systems and equipment and living on board, but excluding the power not for propulsion machinery/systems, e.g. thrusters, cargo pumps, cargo gear, ballast pumps, maintaining cargo, e.g. reefers and cargo hold fans, in the condition where the ship engaged in voyage at the speed (Vref) under the condition as mentioned in paragraph 2.2.

.1 For ships with a main engine power of 10,000 kW or above, PAE is defined as:

250)75 . 0(025 . 0 1
) (
1
) 10000(

nPTI
i
i PTInME
i
MEiKWMCRMEAE
P
MCRP

.2 For ships with a main engine power below 10,000 kW, PAE is defined as:

)75 . 0(05 . 0 1
) (
1
) 10000(
nPTI
i
i PTInME
i
MEiKWMCRMEAE
P
MCRP

RESOLUTION MEPC.212(63)
Adopted on 2 March 2012

MCRP

.2 For ships with a main engine power below 10,000 kW, PAE is defined as:

)75 . 0(05 . 0 1
) (
1
) 10000(
nPTI
i
i PTInME
i
MEiKWMCRMEAE
P
MCRP

.3 For ship where the P AE value calculated by paragraph 2.5.6.1 or 2.5.6.2 is significantly different from the total power used at normal seagoing, e.g. in cases of passenger ships (see NOTE under the formula of EEDI), the PAE value should be estimated by the consumed electric power (excluding propulsion) in conditions when the ship is engaged in a voyage at reference speed (Vref) as given in the electric power table 1, divided by the average efficiency of the generator(s) weighted by power (see appendix 2).

.6 Vref, Capacity and P should be consistent with each other.

.7 SFC is the certified specific fuel consumption, measured in g/kWh, of the engines. The subscripts ME(i) and AE(i) refer to the main and auxiliary engine(s),

(see appendix 2).

.6 Vref, Capacity and P should be consistent with each other.

.7 SFC is the certified specific fuel consumption, measured in g/kWh, of the engines. The subscripts ME(i) and AE(i) refer to the main and auxiliary engine(s), respectively. For engines certified to the E2 or E3 test cycles of the NO x Technical Code 2008, the engine Specific Fuel Consumption (SFCME(i)) is that recorded in the test report included in a NO x technical file for the engine(s) at 75 per cent of MCR power of its torque rating. For engines certified to the D2 or C1 test cycles of the NO x Technical Code 2008, the engine Specific Fuel Consumption (SFCAE(i)) is that recorded on the test report included in a NO x technical file at the engine(s) 50 per cent of MCR power or torque rating.

The SFC should be corrected to the value corresponding to the ISO standard reference conditions using the standard lower calorific value of the fuel oil (42,700kJ/kg), referring to ISO 15550:2002 and ISO 3046-1:2002.

For ships where the PAE value calculated by paragraphs 2.5.6.1 and 2.5.6.2 is significantly different from the total power used at normal seagoing,

fuel oil (42,700kJ/kg), referring to ISO 15550:2002 and ISO 3046-1:2002.

For ships where the PAE value calculated by paragraphs 2.5.6.1 and 2.5.6.2 is significantly different from the total power used at normal seagoing, e.g. conventional passenger ships, the Specific Fuel Consumption (SFCAE) of the auxiliary generators is that recorded in the test report included in a NOx technical file for the engine(s) at 75 per cent of MCR power of its torque rating.

SFCAE is the power-weighted average among SFC AE(i) of the respective engines i.

For those engines which do not have a test report included in a NO x technical file because its power is below 130 kW, the SFC specified by the manufacturer and endorsed by a competent authority should be used.

At the design stage, in case of unavailability of test report in the NO x file, the SFC specified by the manufacturer and endorsed by a competent authority should be used.

For LNG driven engines of which SFC is measured in kJ/kWh should be corrected to the SFC value of g/kWh using the standard lower calorific value of the LNG (48,000 kJ/kg), referring to the 2006 IPCC Guidelines.

authority should be used.

For LNG driven engines of which SFC is measured in kJ/kWh should be corrected to the SFC value of g/kWh using the standard lower calorific value of the LNG (48,000 kJ/kg), referring to the 2006 IPCC Guidelines.

1 The electric power table should be examined and valida ted by the verifier. W here ambient conditions affect any electrical load in the power table the contractual ambient conditions leading to the maximum design electrical load of the installed system for the ship in general should apply.

.8 fj is a correction factor to account for ship specific design elements:

.1 The power correction factor, fj, for ice-classed ships should be taken as the greater value of fj0 and f j,min as tabulated in Table 1 but not greater than fj,max = 1.0.

For further information on approximate correspondence between ice classes, see HELCOM Recommendation 25/72.

taken as the greater value of fj0 and f j,min as tabulated in Table 1 but not greater than fj,max = 1.0.

For further information on approximate correspondence between ice classes, see HELCOM Recommendation 25/72.

Table 1: Correction factor for power f j for ice-classed ships

Ship type	fj0	fj,min depending on the ice class			
		IA Super	IA	IB	IC
Tanker nME i i ME PP P L 1) (	920 . 1 308 . 0 30 . 0 15 . 0PPL 21 . 0 27 . 0PPL	13 . 045 . 0PPL	06 . 070 . 0PPL		
Bulk carrier nME i i ME PP P L 1) (	754 . 1 639 . 0	09 . 0 47 . 0PPL 07 . 0 58 . 0PPL 04 . 0 73 . 0PPL 02 . 0 87 . 0PPL			
General cargo ship nME i i ME PP P L 1					

) (
483 . 2
0227 . 0 16 . 0
31 . 0PPL
12 . 0
43 . 0PPL 09 . 056 . 0PPL 07 . 067 . 0PPL

.2 The factor fj, for shuttle tankers with propulsion redundancy should be fj = 0.77. This correction factors applies to shuttle tankers with propulsion redundancy between 80,000 and 160,000 deadweight. The Shuttle Tankers with Propulsion Redundancy are tankers used for loading of crude oil from offshore installations equipped with dual-engine and twin-propellers need to meet the

propulsion redundancy between 80,000 and 160,000 deadweight. The Shuttle Tankers with Propulsion Redundancy are tankers used for loading of crude oil from offshore installations equipped with dual-engine and twin-propellers need to meet the requirements for dynamic positioning and redundancy propulsion class notation.

.3 For other ship types, fj should be taken as 1.0.

.9 fw is a non-dimensional coefficient indicating the decrease of speed in representative sea conditions of wave height, wave frequency and wind speed (e.g. Beaufort Scale 6), and is determined as follows:

.1 for attained EEDI calculated under regulations 20 and 21 of MARPOL Annex VI, fw is 1.00;

.2 when fw is calculated according to the subparagraph .2.1 or .2.2 below, the value for attained EEDI calculated by the formula in paragraph 2 using the obtained fw should be referred to as "attained EEDIweather";

.1 fw can be determined by conducting the ship specific simulation on its performance at representative sea conditions. The simulation methodology should be based on the Guidelines developed by the Organization and the

simulation on its performance at representative sea conditions. The simulation methodology should be based on the Guidelines developed by the Organization and the

2 HELCOM Recommendation 25/7 may be found at http://www.helcom.fi.

method and outcome for an individual ship should be verified by the Administration or an organization recognized by the Administration; and

.2 in cases where a simulation is not conducted, fw should be taken from the "Standard fw " table/curve. A "Standard fw " table/curve is provided in the Guidelines 3 for each ship type defined in paragraph 1, and expressed as a function of Capacity (e.g. deadweight). The "Standard fw " table/curve is based on data of actual speed reduction of as many existing ships as possible under the representative sea condition.

fw and attained EEDI weather, if calculated, with the representative sea

table/curve is based on data of actual speed reduction of as many existing ships as possible under the representative sea condition.

fw and attained EEDI weather, if calculated, with the representative sea conditions under which those values are determined, should be indicated in the EEDI Technical File to make a distinction with the attained EEDI calculated under regulations 20 and 21 of MARPOL Annex VI.

.10 feff(i) is the availability factor of each innovative energy efficiency technology. feff(i) for waste energy recovery system should be one (1.0)4.

.11 fi is the capacity factor for any tech nical/regulatory limitation on capacity, and should be assumed to be one (1.0) if no necessity of the factor is granted.

.1 The capacity correction factor, fi, for ice-classed ships should be taken as the lesser value of fi0 and fi,max as tabulated in Table 2, but not less than fi,min = 1.0. For further information on approximate correspondence between ice classes, see HELCOM Recommendation 25/75.

Table 2: Capacity correction factor fi for ice-classed ships

Ship type	fi0	fi,max depending on the ice class			
		IA Super	IA	IB	IC
Tanker	capacityLPP				
	331 . 3				

correspondence between ice classes, see HELCOM
Recommendation 25/75.

Table 2: Capacity correction factor fi for ice-classed ships
Ship type fi0
fi,max depending on the ice class
IA Super IA IB IC
Tanker capacityLPP
331 . 3
00138 . 0 11 . 010 . 2PPL 08 . 071 . 1PPL 06 . 047 . 1PPL 04 . 027 . 1PPL
Bulk carrier capacityLPP
123 . 3
00403 . 0 11 . 010 . 2PPL 09 . 080 . 1PPL 07 . 054 . 1PPL 05 . 031 . 1PPL
General
cargo ship capacity
LPP 625 . 20377 . 0 11 . 018 . 2PPL 08 . 077 . 1PPL 06 . 051 . 1PPL 04 . 028 . 1PPL
Containership capacityLPP 329 . 21033 . 0 11 . 010 . 2PPL 08 . 071 . 1PPL 06 . 047 . 1PPL 04 . 027 . 1PPL
Gas carrier capacityLPP 590 . 20474 . 0 25 . 1 12 . 010 . 2PPL 08 . 060 . 1PPL 04 . 025 . 1PPL
Note: containership capacity is defined as 70% of the DWT.

3 Guidelines for the calculation of the coefficient fw for the decrease of ship speed in respective sea conditions will be developed.
4 EEDI calculation should be based on the normal sea-going condition outside Emission Control Area designated under paragraph 6 of regulation 13 in MARPOL ANNEX VI.

conditions will be developed.
4 EEDI calculation should be based on the normal sea-going condition outside Emission Control Area designated under paragraph 6 of regulation 13 in MARPOL ANNEX VI.
5 HELCOM Recommendation 25/7 may be found at http://www.helcom.fi .
RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 11

I:\MEPC\63\23.doc
.2 f i VSE for ship specific voluntary structural enhancement is expressed by the following formula:

designenhanced
designreference
VSEi DWT
DWTf
Where:

designreferenceshipdesignreference t lightweighDWT

designenhancedshipdesignenhanced t lightweighDWT

For this calculation the same displacement ( ) for reference and enhanced design should be taken.

DWT before enhancements (DWTreference design) is the deadweight prior to application of the structural enhancements. DWT after enhancements (DWTenhanced design) is the deadweight following the application of voluntary structural enhancement. A change of material (e.g. from aluminum alloy to steel) between reference

enhancements (DWTenhanced design) is the deadweight following the application of voluntary structural enhancement. A change of material (e.g. from aluminum alloy to steel) between reference design and enhanced design should not be allowed for the fi VSE calculation. A change in grade of the same material (e.g. in steel type, grades, properties and condition) should also not be allowed.

In each case, two sets of structural plans of the ship should be submitted to the verifier for assessment. One set for the ship without voluntary structural enhancement; the other set for the same ship with voluntary structural enhancement. (Alternatively, one set of structural plans of the reference design with annotations of voluntary structural enhancement should also be acceptable.) Both sets of structural plans should comply with the applicable regulations for the ship type and intended trade.

.3 for bulk carriers and oil tankers, built in accordance with Common Structural Rules (CSR) of the classification societies and assigned the class notation CSR, the following capacity correction factor fiCSR should apply:

$$fi_{CSR} = 1 + (0.08 \cdot LWT_{CSR} / DWT_{CSR})$$

Structural Rules (CSR) of the classification societies and assigned the class notation CSR, the following capacity correction factor fiCSR should apply:

fiCSR = 1 + (0.08 · LWTCSR / DWTCSR)
Where, DWTCSR is the deadweight determined by paragraph 2.4 and LWTCSR is the light weight of the ship.

.4 for other ship types, fi should be taken as 1.0.

.12 fc is the cubic capacity correction factor and should be assumed to be one (1.0) if no necessity of the factor is granted.

.1 for chemical tankers, as defined in regulation 1.16.1 of MARPOL Annex II, the following cubic capacity correction factor fc should apply:

fc = R 0.7 0.014, where R is less than 0.98
or
fc = 1.000, where R is 0.98 and above;
RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 12

I:\MEPC\63\23.doc
where: R is the capacity ratio of the deadweight of the ship (tonnes) as determined by paragraph 2.4 divided by the total cubic capacity of the cargo tanks of the ship (m3).

.2 for gas carriers having direct diesel driven propulsion system

where: R is the capacity ratio of the deadweight of the ship (tonnes) as determined by paragraph 2.4 divided by the total cubic capacity of the cargo tanks of the ship (m3).

.2 for gas carriers having direct diesel driven propulsion system constructed or adapted and used for the carriage in bulk of liquefied natural gas, the following cubic capacity correction factor fcLNG should apply:

fcLNG = R -0.56

where, R is capacity ratio of deadweight of the ship (tonnes) as determined by paragraph 2.4 divided by the total cubic capacity of the cargo tanks of the ship (m3).

.13 Length between perpendiculars , Lpp means 96 per cent of the total length on a waterline at 85 per cent of the least moulded depth measured from the top of the keel, or the length from the foreside of the stem to the axis of the rudder stock on that waterline, if that were greater. In ships designed with a rake of keel the waterline on which this length is measured should be parallel to the designed waterline. The length between perpendiculars (L pp) should be measured in metres.

RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE

parallel to the designed waterline. The length between perpendiculars (L pp) should be measured in metres.

APPENDIX 1

A GENERIC AND SIMPLIFIED MARINE POWER PLANT

Note 1: Mechanical recovered waste energy directly coupled to shafts need not be measured, since the effect of the technology is directly reflected in the Vref .

Note 2: In case of combined PTI/PTO, the normal operational mode at sea will determine which of these to be used in the calculation.

AUXILIARY
ENGINES

BOILER

CARGO HEAT

THRUSTERS

CARGO PUMPS

REEFERS

CARGO GEAR

BALLAST PUMPS
SWITCH BOARD
SHAFT MOTOR PPTI

WASTE HEAT
RECOVERY etc.
PAEeff
MAIN ENGINE PME SHAFT POWER
PS
MAIN ENGINE
PUMPS (2.5% PME)
ACCOMMODATION
(250 kW)
PAE
SHAFT
GENERATOR PPTO

ACCOMMODATION
(250 kW)
PAE
SHAFT
GENERATOR PPTO
RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 14

I:\MEPC\63\23.doc
APPENDIX 2

GUIDELINES FOR THE DEVELOPMENT OF ELECTRIC POWER TABLES
FOR EEDI (EPT-EEDI)

1 Introduction to the document "Electric Power Table for EEDI"

1.1 This appendix contains a guideline for the document "Electric Power Table for EEDI" which is similar to the actual shipyards' l oad balance document, utilizing well defined criteria, providing standard format, clear loads definition and grouping, standard load factors, etc. A number of new definitions (in particular the "groups") are introduced, giving an apparent greater complexity to the calculation process. However, this intermediate step to the final calculation of PAE stimulates all the parties to a deep investigation through the global figure of the auxiliary load, allowing comparisons between different ships and technologies and eventually identifying potential efficiencies improvements.

2 Auxiliary load power definition

the auxiliary load, allowing comparisons between different ships and technologies and eventually identifying potential efficiencies improvements.

2 Auxiliary load power definition

2.2 PAE is to be calculated as indicated in paragraph 2.5.6 of the Guidelines, together with the following additional three conditions:

.1 no emergency situations (e.g. "no fire", "no flood", "no blackout", "no partial blackout");

.2 evaluation time frame of 24 hours (to account loads with intermittent use); and

.3 ship fully loaded of passenger and/or cargo and crew.

3 Definition of the data to be included in the Electric Power Table for EEDI

3.1 The Electric power table for EEDI calculation should contain the following data elements, as appropriate:

.1 Load's group;
.2 Load's description;
.3 Load's identification tag;
.4 Load's electric circuit Identification;
.5 Load's mechanical rated power " Pm" [kW];
.6 Load's electric motor rated output power [kW];
.7 Load's electric motor efficiency " e" [/];
.8 Load's Rated electric power " Pr" [kW];
.9 Service factor of load " kl" [/];
.10 Service factor of duty " kd" [/];
.11 Service factor of time " kt" [/];

.7 Load's electric motor efficiency " e" [/];
.8 Load's Rated electric power " Pr" [kW];
.9 Service factor of load " kl" [/];
.10 Service factor of duty " kd" [/];
.11 Service factor of time " kt" [/];
.12 Service total factor of use " ku" [/], where ku=kl·kd·kt;
.13 Load's necessary power "Pload" [kW], where Pload=Pr·ku;
.14 Notes;
.15 Group's necessary power [kW]; and
.16 Auxiliaries load's power P AE [kW].

RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 15

I:\MEPC\63\23.doc
4 Data to be included in the Electric Power Table for EEDI

Load groups

4.1 The Loads are put into defined groups, allowing a proper breakdown of the auxiliaries. This eases the verification process and makes it possible to identify those areas where load reductions might be possible. The groups are listed below:

.1 A Hull, Deck, Navigation and Safety services;
.2 B Propulsion service auxiliaries;
.3 C Auxiliary Engine and Main Engine Services;
.4 D Ship's General services;

where load reductions might be possible. The groups are listed below:

.1 A Hull, Deck, Navigation and Safety services;
.2 B Propulsion service auxiliaries;
.3 C Auxiliary Engine and Main Engine Services;
.4 D Ship's General services;
.5 E Ventilation for Engine-rooms and Auxiliaries room;
.6 F Air Conditioning services;
.7 G Galleys, refrigeration and Laundries services;
.8 H Accommodation services;
.9 I Lighting and socket services;
.10 L Entertainment services;
.11 N Cargo loads; and
.12 M Miscellaneous.

All the ship's loads have to be delineated in the document, excluding only PAeff, the shaft motors and shaft motors chain (while the propulsion services auxiliaries are partially included below in paragraph 4.1.2 B). Some loads (i.e. thrusters, cargo pumps, cargo gear, ballast pumps, maintaining cargo, reefers and cargo hold fans) still are included in the group for sake of transparency, however their service factor is zero in order to comply with rows 4 and 5 of paragraph 2.5.6 of the Guidelines, therefore making it easier to verify that all the loads have been considered in the document and there are no loads left out of the

and 5 of paragraph 2.5.6 of the Guidelines, therefore making it easier to verify that all the loads have been considered in the document and there are no loads left out of the measurement.

4.1.1 A Hull, Deck, Navigation and safety services

.1 loads included in the Hull services typically are: ICCP systems, mooring equipment, various doors, ballasting systems, Bilge systems, Stabilizing equipment, etc. Ballasting systems are indicated with service factor equal to zero to comply with row 5 of paragraph 2.5.6 of the Guidelines;

.2 loads included in the deck services typically are: deck and balcony washing systems, rescue systems, cranes, etc.;

.3 loads included in the navigation servic es typically are: navigation systems, navigation's external and internal communication systems, steering systems, etc.; and

.4 loads included in the safety services typically are: active and passive fire systems, emergency shutdown systems, public address systems, etc.

4.1.2 B Propulsion service auxiliaries
This group typically includes: propulsion secon dary cooling systems such as LT cooling

systems, emergency shutdown systems, public address systems, etc.

4.1.2 B Propulsion service auxiliaries
This group typically includes: propulsion secon dary cooling systems such as LT cooling pumps dedicated to shaft motors, LT cooling pumps dedicated to propulsion converters, propulsion UPSs, etc. Propulsion service Loads do not include shaft motors (PTI(i)) and the auxiliaries which are part of them (shaft motor own cooling fans and pump, etc.) and the shaft motor chain losses and auxiliaries which are part of them (i.e. shaft motor converters including relevant auxiliaries such as converter own cooling fans and pumps, shaft motor

transformers including relevant auxiliaries losses such as propulsion transformer own cooling fans and pumps, shaft motor Harmonic filter including relevant auxiliaries losses, shaft motor excitation system including the relevant auxiliaries consumed power, etc.). Propulsion

fans and pumps, shaft motor Harmonic filter including relevant auxiliaries losses, shaft motor excitation system including the relevant auxiliaries consumed power, etc.). Propulsion service auxiliaries include manoeuvring propul sion equipments such as manoeuvring thrusters and their auxiliaries whose service factor is to be set to zero.

4.1.3 C Auxiliary Engine and Main Engine Services
This group includes: cooling systems, i.e. pumps and fans for cooling circuits dedicated to alternators or propulsion shaft engines (seawater, technical water dedicated pumps, etc.), lubricating and fuel systems feeding, transfer, treatment and storage, ventilation system for combustion air supply, etc.

4.1.4 D Ship's General services
This group includes Loads which provide general services which can be shared between shaft motor, auxiliary engines and main engine and accommodation support systems. Loads typically included in this group are: Cooling systems, i.e. pumping seawater, technical water main circuits, compressed air systems, fresh water generators, automation systems, etc.

4.1.5 E Ventilation for Engine-rooms and Auxiliaries room

typically included in this group are: Cooling systems, i.e. pumping seawater, technical water main circuits, compressed air systems, fresh water generators, automation systems, etc.

4.1.5 E Ventilation for Engine-rooms and Auxiliaries room
This group includes all fans providing ventila tion for engine-rooms and auxiliary rooms that typically are: Engine-rooms cooling supply-ex haust fans, auxiliary rooms supply and exhaust fans. All the fans serving accommodation areas or supplying combustion air are not included in this group. This group does not include cargo hold fans, and garage supply and exhaust fans.

4.1.6 F Air Conditioning services
All Loads that make up the air conditioning service that typically are: air conditioning chillers, air conditioning cooling and heating fluids transfer and treatment, air conditioning's air handling units ventilation, air conditioning re-heating systems with associated pumping, etc. The air conditioning chillers service factor of load, service factor of time and service factor of duty are to be set as 1 (kl=1, kt=1 and kd=1) in order to avoid the detailed validation of the

The air conditioning chillers service factor of load, service factor of time and service factor of duty are to be set as 1 (kl=1, kt=1 and kd=1) in order to avoid the detailed validation of the heat load dissipation document (i.e. the chiller's electric motor rated power is to be used). However, kd is to represent the use of spare chillers (e.g. four chillers are installed and one out four is spare then kd=0 for the spare chiller and kd=1 for the remaining three chillers), but only when the number of spare chillers is cl early demonstrated via the heat load dissipation document.

4.1.7 G Galleys, refrigeration and Laundries services
All Loads related to the galleys, pantries refrigeration and laundry services that typically are: Galleys various machines, cooking appliances, galleys' cleaning machines, galleys auxiliaries, refrigerated room systems including refrigeration compressors with auxiliaries, air coolers, etc.

4.1.8 H Accommodation services
All Loads related to the accommodation services of passengers and crew that typically are: crew and passengers' transportation systems, i. e. lifts, escalators, etc., environmental

coolers, etc.

4.1.8 H Accommodation services
All Loads related to the accommodation services of passengers and crew that typically are: crew and passengers' transportation systems, i. e. lifts, escalators, etc., environmental services, i.e. black and grey water collecting, transfer, treatment, storage, discharge, waste systems including collecting, transfer, treat ment, storage, etc., accommodation fluids transfers, i.e. sanitary hot and cold water pu mping, etc., treatment units, pools systems, saunas, gym equipments, etc.

4.1.9 I Lighting and socket services
All Loads related to the lighting, entertainment and socket services. As the quantity of lighting circuits and sockets within the ship may be significantly high, it is not practically feasible to list all the lighting circuits and points in the EPT for EEDI. Therefore circuits should be grouped into subgroups aimed to identify possible improvements of efficient use of

feasible to list all the lighting circuits and points in the EPT for EEDI. Therefore circuits should be grouped into subgroups aimed to identify possible improvements of efficient use of power. The subgroups are:

.1 Lighting for 1) cabins, 2) corridors, 3) technical rooms/stairs, 4) public spaces/stairs, 5) engine-rooms and auxiliaries' room, 6) external areas, 7) garages and 8) cargo spaces. All have to be divided by main vertical zone; and

.2 Power sockets for 1) cabins, 2) corridors, 3) technical rooms/stairs, 4) public spaces/stairs, 5) engine-rooms and auxiliaries' room, 6) garages and 7) cargo spaces. All have to be divided by main vertical zone.

The calculation criteria for complex groups (e.g. cabin lighting and power sockets) subgroups are to be included via an explanatory note, indicating the load composition (e.g. lights of typical cabins, TV, hair dryer, fridge, etc., typical cabins).

4.1.10 L Entertainment services

This group includes all Loads related to the enterta inment services that typically are: public spaces audio and video equipments, theatre stage equipments, IT systems for offices, video games, etc.

4.1.11 N Cargo Loads

This group includes all Loads related to the enterta inment services that typically are: public spaces audio and video equipments, theatre stage equipments, IT systems for offices, video games, etc.

4.1.11 N Cargo Loads
This group will contain all cargo loads such as cargo pumps, cargo gear, maintaining cargo, cargo reefers loads, cargo hold fans and garage fans for sake of transparency. However, the service factor of this group is to be set to zero.

4.1.12 M Miscellaneous
This group will contain all loads which have not been associated to the above-mentioned groups but still are contributing to the overall load calculation of the normal maximum sea load.

Loads description

4.2 This identifies the loads (for example "seawater pump").

Loads identification tag

4.3 This tag identifies the loads according to the shipyard's standards tagging system. For example, the "PTI1 fresh water pump" identification tag is "SYYIA/C" for an example ship and shipyard. This data provides a unique identifier for each load.

Loads electric circuit Identification

4.4 This is the tag of the electric circuit supplying the load. Such information allows the

and shipyard. This data provides a unique identifier for each load.

Loads electric circuit Identification

4.4 This is the tag of the electric circuit supplying the load. Such information allows the data validation process.

Loads mechanical rated power "Pm"

4.5 This data is to be indicated in the document only when th electric load is made by an electric motor driving a mechanical load (for example a fan, a pump, etc.). This is the rated power of the mechanical device driven by an electric motor.

Loads electric motor rated output power [kW]

4.6 The output power of the electric motor as per maker's name plate or technical specification. This data does not take part of the calculation but is useful to highlight potential over rating of the combination motor-mechanical load.

Loads electric motor efficiency "e" [/]

4.7 This data is to be entered in the document only when the electric load is made by an

potential over rating of the combination motor-mechanical load.

Loads electric motor efficiency "e" [/]

4.7 This data is to be entered in the document only when the electric load is made by an electric motor driving a mechanical load.

Loads rated electric power "Pr" [kW]

4.8 Typically the maximum electric power abs orbed at the load electric terminals at which the load has been designed for its service, as indicated on the maker's name plate and/or maker's technical specification. When t he electric load is made by an electric motor driving a mechanical load the load's rated electric power is: Pr=Pm/e [kW].

Service factor of load "kl" [/]

4.9 Provides the reduction from the loads rated electric power to loads necessary electric power that is to be made when the load absorb less power than its rated power. For example, in case of electric motor driving a mechanical load, a fan could be designed with some power margin, leading to the fact that the fan rated mechanical power exceeds the power requested by the duct system it serves. Another example is when a pump rated

with some power margin, leading to the fact that the fan rated mechanical power exceeds the power requested by the duct system it serves. Another example is when a pump rated power exceed the power needed for pumping in its de livery fluid circuit. Another example in case of electric self-regulating semi-conductors electric heating system is oversized and the rated power exceeds the power absorbed, according a factor kl.

Service factor of duty "kd" [/]

4.10 Factor of duty is to be used when a function is provided by more than one load. As all loads have to be included in the EPT for EEDI, this factor provides a correct summation of the loads. For example when two pumps serve the same circuit and they run in duty/stand-by their Kd factor will be ½ and ½. When three compressors serves the same circuit and one runs in duty and two in stand-by, then kd is 1/3, 1/3 and 1/3.

Service factor of time "kt" [/]

4.11 A factor of time based on the shipyard's evaluation about the load duty along 24 hours of ship's navigation as defined at paragraph 3. For example the Entertainment loads operate

Service factor of time "kt" [/]

4.11 A factor of time based on the shipyard's evaluation about the load duty along 24 hours of ship's navigation as defined at paragraph 3. For example the Entertainment loads operate at their power for a limited period of time, 4 hours out 24 hours; as a consequence kt=4/24. For example, the seawater cooling pumps operate at their power all the time during the navigation at Vref. As a consequence kt=1.

Service total factor of use "ku" [/]

4.12 The total factor of use that takes into consideration all the service factors: ku=kl·kd·kt.

Loads necessary power "Pload" [kW]

4.13 The individual user contribution to the auxiliary load power is Pload=Pr·ku.

Notes

4.14 A note, as free text, could be included in the document to provide explanations to the verifier.

Groups necessary power [kW]

4.15 The summation of the "Loads necessary power" from group A to N. This is an

Notes

4.14 A note, as free text, could be included in the document to provide explanations to the verifier.

Groups necessary power [kW]

4.15 The summation of the "Loads necessary power" from group A to N. This is an intermediate step which is not strictly necessary for the calculation of PAE. However, it is useful to allow a quantitative analysis of the PAE, providing a standard breakdown for analysis and potential improvements of energy saving.

Auxiliaries load's power PAE [kW]

4.16 Auxiliaries load's power PAE is the summation of the "Load's necessary power" of all the loads divided by the average efficiency of the generator(s) weighted by power.

PAE=Pload(i)/(average efficiency of the generator(s) weighted by power)

Layout and organization of the data indicated in the "Electric power table for EEDI"

5 The document "Electric power table for EEDI" is to include general information (i.e. ship's name, project name, document references, etc.) and a table with:

.1 one row containing column titles;

.2 one Column for table row ID;

.3 one Column for the groups identification ("A", "B", etc.) as indicated in

(i.e. ship's name, project name, document references, etc.) and a table with:

.1 one row containing column titles;

.2 one Column for table row ID;

.3 one Column for the groups identification ("A", "B", etc.) as indicated in paragraphs 4.1.1 to 4.1.12 of this guideline;

.4 one Column for the group descriptions as indicated in paragraphs 4.1.1 to 4.1.12 of this guideline;

.5 one column each for items in paragraphs 4.2 to 4.14 of this guideline (e.g. "load tag", etc.);

.6 one row dedicated to each individual load;

.7 the summation results (i.e. summation of powers) including data from paragraphs 4.15 to 4.16 of this guideline; and

.8 explanatory notes.

An example of an Electric Power Table for EEDI for a cruise postal vessel which transports passenger and have a car garage and reefer holds for fish trade transportation is indicated below. The data indicated and the type of ship is for reference only.

RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 20

I:\MEPC\63\23.doc

***

Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 20

I:\MEPC\63\23.doc

***

RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSRESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPS